

HERO scorecard (vertical) — value ± BCa 95% CI · colour = per-metric min-max (green = best) · \* = Δ-vs-frozen CI excludes 0  
9 metrics × 12 encoders · WINNER col + BLUE BOX = surgery-vs-vanilla-cont-SSL champion duel (tie = paired 95% CI overlaps 0) · 10 K bootstrap

|                       | G-frozen         | G-fullFT          | G-LPFT            | G-DoRA            | G-LoRA            | G-vCSSL-enc       | G-raw             | G-s3DI-enc        | G-s3DI-hd         | G-argn           | G-sNoDI-enc       | G-sNoDI-hd        | WINNER                            |
|-----------------------|------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|------------------|-------------------|-------------------|-----------------------------------|
| action_top1 ↑ better  | 44.274<br>±2.274 | 52.712<br>±2.301* | 49.041<br>±2.329* | 52.603<br>±2.329* | 50.904<br>±2.274* | 52.822<br>±2.301* | 50.685<br>±2.301* | 49.151<br>±2.320* | 49.973<br>±2.301* | N/A              | 50.575<br>±2.301* | 49.644<br>±2.274* | ViT-G vanilla<br>continual-SSL    |
| motion_cos ↑ better   | 0.009<br>±0.001  | 0.244<br>±0.006*  | 0.119<br>±0.004*  | 0.161<br>±0.005*  | 0.170<br>±0.005*  | 0.151<br>±0.004*  | 0.133<br>±0.004*  | 0.119<br>±0.004*  | 0.127<br>±0.005*  | 0.189<br>±0.005* | 0.127<br>±0.004*  | 0.135<br>±0.005*  | ViT-G surgical<br>autorgn encoder |
| taxonomy_f1 ↑ better  | 0.797<br>±0.017  | 0.785<br>±0.017   | 0.789<br>±0.017   | 0.782<br>±0.017   | 0.786<br>±0.017   | 0.790<br>±0.017   | 0.792<br>±0.017   | 0.789<br>±0.017   | 0.787<br>±0.017   | N/A              | 0.785<br>±0.017   | 0.787<br>±0.017   | tie                               |
| future_mse ↓ better   | 0.557<br>±0.001  | 0.541<br>±0.001*  | 0.536<br>±0.001*  | 0.514<br>±0.001*  | 0.504<br>±0.001*  | 0.533<br>±0.001*  | 0.497<br>±0.001*  | 0.498<br>±0.001*  | 0.533<br>±0.001*  | 0.526<br>±0.001* | 0.498<br>±0.001*  | 0.533<br>±0.001*  | ViT-G surgery<br>raw encoder      |
| rollout ↓ better      | 0.009<br>±0.000  | 0.010<br>±0.000*  | 0.010<br>±0.000*  | 0.011<br>±0.000*  | 0.009<br>±0.000*  | 0.008<br>±0.000*  | 0.010<br>±0.000*  | 0.010<br>±0.000*  | 0.008<br>±0.000*  | 0.009<br>±0.000* | 0.012<br>±0.000*  | 0.008<br>±0.000*  | tie                               |
| causal ↓ better       | 0.583<br>±0.001  | 0.566<br>±0.001*  | 0.561<br>±0.001*  | 0.542<br>±0.001*  | 0.533<br>±0.001*  | 0.558<br>±0.001*  | 0.528<br>±0.002*  | 0.530<br>±0.001*  | 0.558<br>±0.001*  | 0.551<br>±0.001* | 0.530<br>±0.002*  | 0.558<br>±0.001*  | ViT-G surgery<br>raw encoder      |
| tdist ↓ better        | 0.007<br>±0.000  | 0.008<br>±0.000*  | 0.008<br>±0.000*  | 0.007<br>±0.000*  | 0.008<br>±0.000*  | 0.006<br>±0.000*  | 0.008<br>±0.000*  | 0.007<br>±0.000*  | 0.006<br>±0.000*  | 0.008<br>±0.000* | 0.008<br>±0.000*  | 0.006<br>±0.000*  | tie                               |
| teacher_free ↓ better | 0.046<br>±0.000  | 0.049<br>±0.001*  | 0.048<br>±0.001*  | 0.051<br>±0.001*  | 0.045<br>±0.001   | 0.038<br>±0.001*  | 0.045<br>±0.001*  | 0.045<br>±0.001   | 0.038<br>±0.001*  | 0.047<br>±0.001* | 0.052<br>±0.001*  | 0.038<br>±0.001*  | tie                               |
| maskratio ↓ better    | 0.072<br>±0.001  | 0.066<br>±0.001*  | 0.061<br>±0.001*  | 0.057<br>±0.000*  | 0.054<br>±0.000*  | 0.063<br>±0.001*  | 0.056<br>±0.001*  | 0.057<br>±0.000*  | 0.063<br>±0.001*  | 0.061<br>±0.001* | 0.057<br>±0.001*  | 0.063<br>±0.001*  | ViT-G surgery<br>raw encoder      |